

3. What is detergent?

- (a) Soap (b) Medicine
(c) Catalyst (d) Refining agent

40th B.P.S.C. (Pre) 1995

Ans. (d)

A detergent is different from soap. A detergent is a surfactant or a mixture of surfactants with cleaning and refining properties in dilute solutions. The most important ingredients in detergents are chemicals called surfactants. The term detergent by itself refers specifically to laundry detergent or dish detergent as opposed to hand soap or other types of cleaning agents. Detergents are commonly available as a powder or concentrated solutions.

4. Detergents used for cleaning clothes are :

- (a) Carbonates (b) Bicarbonates
(c) Bismuthates (d) Sulfonates
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Detergents used for cleaning clothes are usually alkylbenzene sulfonates, a family of compounds that are similar to soap but are more soluble in hard water.

5. Assertion (A) : Detergent easily removes oil and dirt from dirty garments.

Reason (R) : Detergent increases the surface tension of water.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

U.P.P.C.S. (Pre) 1997

Ans. (c)

Assertion (A) is true as the detergent easily removes oil and dirt from dirty garments. The surface tension of water is dramatically reduced by the addition of soap or detergent. The soap or detergent molecules tend to aggregate at the surface of the water. By this, they weaken the hydrogen bonding between water molecules, hence, surface tension reduces. Hence (A) is true, but (R) is false.

6. Assertion (A) : Synthetic detergents can make more lather in hard water.

Reason (R) : Synthetic detergents form soluble calcium and magnesium salts with hard water.

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

I.A.S. (Pre) 2002

Ans. (a)

Detergents are different from soap. They are used to clean clothes with hard water as detergent does not form any insoluble precipitate with calcium and magnesium salts. They form water-soluble calcium and magnesium salts. This is the reason why detergents produce more foam with hard water and remove oil and dust from dirty clothes.

7. Consider the following statements :

Sodium silicate and sodium sulphate are mixed as detergents in washing powders so that–

1. To maintain dryness of the powder.

2. To maintain the basicity of powder.

Which of the following statement is correct?

- (a) only 1 (b) only 2
(c) both 1 & 2 (d) None of both

I.A.S. (Pre) 2003

Ans. (c)

By adding sodium sulphate to the washing powder, it increases the basicity of the powder. The addition of sodium silicate into the washing powder turns the powder crystalline, so that the powder stays in dry form.

The Fertilizers

Notes

Fertilizers :

- Fertilizer is any material of natural or synthetic origin that is applied to soils or to plant tissues to supply one or more plant nutrients essential to the growth of plants.
- Fertilizers are used daily by farmers and families to help crops and gardens grow.
- With the global population steadily growing, it is necessary that enough crops are produced each year to provide adequate food, clothing and other agricultural products to people around the world. These crops deplete soil nutrients in different ways and rates.
- Fertilizers play an important role in providing crops with the nutrients they need to grow and be harvested for nutritious food.

Manure :

- Manure is an organic material that is used to fertilize land, usually consisting of the feces and urine of domestic animals, with or without accompanying litter such as straw, hay, etc.

- Animals manure is less rich in nitrogen, phosphorus and potash than **synthetic fertilizers** and hence must be applied in much greater quantities than the latter.
- Manure is rich in organic matter or **humus** and thus improves the soil's capacity to absorb water and store water, thus preventing erosion.
- Much of the potassium and nitrogen in manure can be lost through leaching, if the material is exposed to rainfall before being applied to the field. These nutrient losses may be prevented by such methods as stacking manure under cover or in pits to prevent leaching.
- The manure is also called **compost** or **organic fertilizer**.
- **Green manure** is a type of cover crop, grown to add nutrients and organic matter to the soil. Typically, a green manure crop is grown for a specific period of time, and then ploughed under and incorporated into the soil while it is green or shortly after flowering.
- The **legumes** (mainly pea family plants) used as green manuring crops have nitrogen fixing symbiotic bacteria in their root nodules that fix atmospheric nitrogen in a form that plant can use as well as providing organic matter (humus) to the soil. Such plants are **clover** and **vetch**.
- Continuous use of chemical fertilizers on soil depletes the soil of essential nutrients. As a result, the food produced in these soils have less vitamins and chemical contents. The number of microorganisms in soil may decrease. It also increases the acidity of soil. So the application of organic manure has been recommended along with artificial manure.

Vermicompost :

- Now a days vermicompost is widely used instead of fertilizers. The most significant benefit of vermicompost is that the nutrients in it are very easily absorbed by the roots of plants unlike chemical fertilizers. Vermicompost is not easily flushed from the soil because of the worm mucus that it contains. Plants have longer time to obtain the nutrients and get the maximum benefit.

Difference between Synthetic Fertilizer and Manure

S.N.	Synthetic Fertilizer	Manure
1.	Synthetic fertilizer is an inorganic salt.	Manure is a natural fertilizer.
2.	Synthetic fertilizer is manufactured in factories.	Manure can be made in fields.
3.	Soil does not get humus from the synthetic fertilizer.	Soil gets humus in excessive amount from the manure.
4.	In synthetic fertilizer the plants nutrients as nitrogen, phosphorus and potash are present in abundance amount.	In manure plant nutrients are in less quantity than synthetic fertilizer.

Kinds of Synthetic Fertilizers :

- There are three categories of synthetic fertilizers :
 - (1) Nitrogenous fertilizers
 - (2) Phosphatic fertilizers
 - (3) Potassium fertilizers
- (1) **Nitrogenous fertilizers :**
 - Nitrogen is absorbed by the plant roots in two forms— nitrate form (NO_3^-) and ammonium form (NH_4^+).
 - Most of the crop plants prefer nitrogen in nitrate form; but paddy and few other plants prefer nitrogen in ammonium form. Ammonium form of nitrogen is however, easily convertible into nitrate form.
 - Based on the forms of nitrogen they contain, nitrogenous fertilizers are classified into four categories, viz. (i) nitrate fertilizers (ii) ammonium fertilizers (iii) ammonium nitrate fertilizers and (iv) amide fertilizers.
- (i) **Nitrate fertilizers :** These are three main nitrate fertilizers—
 - (a) Sodium nitrate – 16% N
 - (b) Calcium nitrate – 15.5% N
 - (c) Potassium nitrate – 13% N
- (ii) **Ammonium fertilizers :** The following ammonium fertilizers are in use—
 - (a) Ammonium sulphate – 20% N
 - (b) Ammonium chloride – 24.26% N
 - (c) Monoammonium phosphate – About 11–20% N and 20–48% Phosphate
 - (d) Diammonium phosphate (DAP) – 18% N and 46% phosphate
- (iii) **Ammonium nitrate fertilizers :** Following fertilizers are of this category—
 - (a) Ammonium nitrate – 33% N
 - (b) Calcium ammonium nitrate – 26% N
 - (c) Ammonium sulphate nitrate – 26% N
- (iv) **Amide fertilizers :** Amide group fertilizers are agronomically very significant. They are readily soluble in water and easily decomposed by microorganisms in the soil. In the soil, they are quickly changed into ammonium form and then nitrate form and become available to the plants. The two main amide fertilizers are used –
 - (a) Urea – 46-47% N
 - (b) Calcium cyanamide – 21% N
- (2) **Phosphatic Fertilizers :**
 - The nutrient phosphorus present in phosphate fertilizers are usually expressed in terms of phosphoric anhydride or simply as phosphorus pentoxide (P_2O_5).
 - Some phosphatic fertilizers are as follows :

(a) Super phosphate	–	16-47% P
(b) Basic slag	–	15-25% P
(c) Bone meal	–	20-35% P

- (d) Rock phosphate – 25-30% P
 (e) Calcium monophosphate – 50% P
 (f) Calcium metaphosphate – 64% P
 (g) Ammophos – 46-52% P

(3) Potassium fertilizers :

- It is also called **potash fertilizer**.
- Potassium (K) is one of the essential nutrients for plants, along with nitrogen and phosphorus.
- Potassium is vital for a plant to resist drought and disease as well as for promoting healthy growth.
- The two main types of potash are - Muriate of Potash (MOP) and Sulphate of Potash (SOP).
- **Muriate** of potash is the trade name of potassium chloride.
- Although the amount of potassium in soil is sufficient and no need is required of potassium fertilizer, but its regular use maintains the stability of potassium in soil.
- Some potassium fertilizers are as follows :
 (a) Potassium chloride – 48-60% K
 (b) Potassium sulphate – 48-50% K
 (c) Potassium nitrate – 44% K
 (d) Wood ash – 3.7% K

Biofertilizers :

- Biofertilizers are living microorganisms of bacterial, fungal and algal origin.
- Biofertilizers add nutrients through the natural processes of nitrogen fixation and solubilizing phosphorus and stimulate plant growth through the synthesis of growth-promoting substances.
- Biofertilizers are more cost-effective as compared to chemical fertilizers.
- The microorganisms in biofertilizers restore the soil's natural nutrient cycle and build soil organic matter and they can be expected to reduce the use of synthetic fertilizers and pesticides.

Question Bank

1. **Assertion (A) : Chemical fertilizers are necessary for the phosphorus need of plants.**

Reason (R) : One metric ton of organic manure can barely supply 2 or 3 kg of phosphorus.

From the above statement, which are is correct :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (c) (A) is true, but (R) is false.
 (d) (A) is false, but (R) is true.

U.P. Lower Sub. (Pre) 1998

Ans. (a)

The fertilizer is a substance added to soil to improve plant growth and yield. Nitrogen, Phosphorus and Potassium (N,P,K) are necessary elements of plants. Only 1.5 to 3 kg of Phosphorus is obtained from 1 metric ton of an organic manure. Thus, chemical fertilizers are necessary for the phosphorus need of plants.

2. **For wheat cultivation which fertilizer is used?**

- (a) Nitrogen (b) Potassium
 (c) Copper (d) Iron

47th B.P.S.C. (Pre) 2005

Ans. (a)

Nitrogen is an essential element of plants. For wheat cultivation, the nitrogen fertilizer is most important for its growth and development. The deficiency of nitrogen is probably the most common nutritional problem affecting the plants.

3. **The element not provided by the fertilizers is –**

- (a) Nitrogen (b) Hydrogen
 (c) Chlorine (d) Phosphorus

U.P.P.C.S. (Mains) 2007

Ans. (b)

The fertilizer is a chemical or natural substance added to the soil to increase its fertility. There are at least 16 elements essential for the growth of the plant. Carbon, Hydrogen and Oxygen are derived from carbon dioxide and water. Nitrogen, Phosphorous, Potassium, Calcium, Magnesium, Boron, Chlorine, Copper, Iron, Manganese, Molybdenum and Zinc are normally derived from the soil in the form of inorganic salts.

4. **Which one of the following is not correctly matched?**

Plant Nutrients	Role
(a) Nitrogen	– Growth and Development
(b) Phosphorus	– Root development
(c) Potassium	– Regulating stomatal opening and closing
(d) Boron	– Disease resistance

U.P. R.O./A.R.O. (Pre) 2016

Ans. (*)

Nitrogen, Phosphorus and Potassium are macro-elements of the plants. Nitrogen deficiency causes the yellowing of leaves and retarded growth and development of plants. Plants absorb phosphorus as phosphate ions. It favours the healthy root growth by helping translocation of food. Potassium plays an important role in the opening and closing of stomata. Boron is a micro-element of plants. Its deficiency causes heart rot of beet and internal cork disease in Apple. Its deficiency affects the formation of root nodules in leguminous plants. Boron is also helpful in developing disease resistance power in plants. So, all of the above options are correctly matched.

5. Which one of the following fertilizers leaves maximum acidity in the soil ?

- (a) Urea
- (b) Ammonium Sulphate
- (c) Ammonium Nitrate
- (d) Calcium Ammonium Nitrate

U.P.P.C.S. (Pre) 1996

Ans. (b)

The soil acidic causes a reduction in the yields of acid-sensitive crops. The acidic soils are readily identified by a soil test showing a low pH value. The soil reaction is alkaline when the pH value is above 7, neutral at 7 and acidic below 7. The extensive use of ammonium sulphate is not recommended on acidic soils because of its greater acidifying properties compared to other nitrogen fertilizers.

6. Urea is

- (a) A nitrogen containing organic compound
- (b) A nitrogen containing inorganic compound
- (c) A plant hormone
- (d) An Endergonic compound

U.P.P.C.S. (Pre) 2017

Ans. (a)

Urea is an organic compound whose chemical formula is $\text{CO}(\text{NH}_2)_2$. Nearly 46.7% of nitrogen in the urea is present in the form of amide. In the field of organic chemicals it is also called carbamide.

7. The percentage of nitrogen in urea is –

- (a) 30%
- (b) 25%
- (c) 47%
- (d) 60%

U.P. Lower Sub. (Pre) 2009

Ans. (c)

Urea or Carbamide is an organic compound with a chemical formula $\text{CO}(\text{NH}_2)_2$ having 46-47% nitrogen approximately. The urea molecule has two amide (NH_2) groups containing nitrogen.

8. The elements present in urea are :

- (a) C, H, O
- (b) C, N, O
- (c) C, H, N
- (d) C, H, N, O
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

See the explanation of above question.

9. Which form of nitrogen is available in urea ?

- (a) Amide
- (b) Ammoniacal
- (c) Nitrate
- (d) Nitrite

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (a)

See the explanation of above question.

10. Which of the following chemical fertilizers is relatively more soluble in water at 20°C?

- (a) Ammonium Sulphate
- (b) Urea
- (c) Diammonium phosphate
- (d) Ammonium Chloride

U.P. P.C.S. (Mains) 2017

Ans. (b)

Urea is an organic compound that is only stable in an acidic environment. It is named after the Greek word 'Oura' meaning urine. Urea is very easily soluble in water as compared to other fertilizers.

11. How much amount of urea will be required for supplying 1 kg of nitrogen?

- (a) 2.0 kg
- (b) 2.2 kg
- (c) 2.5 kg
- (d) 2.7 kg

U.P. B.E.O. (Pre) 2019

Ans. (b)

We know that amount of nitrogen in urea = about 46%
 $\therefore$ Urea required for supplying 46 gm of nitrogen = 100 gm
 $\therefore$ Urea required for supplying 1000 gm (1 kg) of nitrogen

$$= \frac{100}{46} \times 1000$$

$$= 2174 \text{ gm}$$

$$= 2.2 \text{ kg (approx.)}$$

Hence, about 2.2 kg of urea will be required for supplying 1 kg of nitrogen.

12. With reference to chemical fertilizers in India, consider the following statements :

1. At present, the retail price of chemical fertilizers is market-driven and not administered by the Government.
2. Ammonia, which is an input of urea, is produced from natural gas.
3. Sulphur, which is a raw material for phosphoric acid fertilizer, is a by-product of oil refineries.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (b)

Statement 1 is incorrect as urea is the main chemical fertilizer used in India, and MRP of urea is statutorily fixed by the Government. Government has implemented the Nutrient Based Subsidy (NBS) policy w.e.f. 1.4.2010 for Phosphatic and Potassic

(P&K) fertilizers. Under the policy, MRP of P&K fertilizers has been left open and fertilizer manufacturers/marketers are allowed to fix the MRP at reasonable rates as per market dynamics, while the Government announces nutrient based subsidy for them.

Ammonia is used to produce various nitrogenous fertilizers, including urea. For the industrial purposes, ammonia is manufactured by Haber's process, for which hydrogen is obtained from natural gas. Natural gas contains methane (CH_4) which is used to produce hydrogen with thermal processes, such as steam-methane reformation and partial oxidation. Hence, statement 2 is correct.

Statement 3 is also correct as in India the domestic production of elemental sulphur is limited to by-product recoveries from petroleum refineries which is a raw material for phosphoric acid fertilizers.

13. What can be the impact of the excessive/ inappropriate use of nitrogenous fertilizers in agriculture?

1. The proliferation of nitrogen-fixing microorganisms in soil can occur.

2. The increase in the acidity of soil can take place.

3. Leaching of nitrate to the groundwater can occur.

Select the correct answer using the code given below.

- (a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2015

Ans. (c)

The excessive/inappropriate use of nitrogenous fertilizers in agriculture can affect environment in various manners for example, increment of acidity in the soil, leaching of ground water, eutrophication etc. The proliferation of nitrogen-fixing microorganisms in soil does not occur due to this. Hence, option (c) is the correct answer.

14. Which of the following is consumed in the manufacture of urea fertilizer?

- (a) Carbon dioxide (b) Sulphur dioxide
(c) Chlorine (d) Nitric acid

Uttarakhand P.C.S. (Mains) 2006

Ans. (a)

The Urea or Carbamide $\text{CO}(\text{NH}_2)_2$ is an organic compound. It is colourless, odourless, highly soluble in water and practically non-toxic. For use in industry, urea is produced from synthetic ammonia and carbon dioxide.

15. Why does the Government of India promote the use of 'Neem-coated Urea' in agriculture?

- (a) Release of Neem oil in the soil increases nitrogen fixation by the soil microorganisms
(b) Neem coating slows down the rate of dissolution of urea in the soil

(c) Nitrous oxide, which is a greenhouse gas, is not at all released into atmosphere by crop fields

(d) It is a combination of a weedicide and a fertilizer for particular crops

I.A.S. (Pre) 2016

Ans. (b)

When urea is applied to soil, it is first hydrolysed (i.e. broken by water) into ammonical form (NH_4^+) and then to nitrite (NO_2^-) followed by nitrate (NO_3^-) formed by the process called nitrification. Most crops use nitrate as a source of nitrogen. However, if the process of nitrification is too rapid, the nitrogen will escape to the atmosphere and plants will not be able to recover it from urea efficiently. So, there is a need to regulate the urea hydrolysis and nitrification. This is done by some chemical/natural agents which are called nitrification inhibitors. It has been scientifically established that the neem oil serves as an effective inhibitor if coated on urea. Neem coating leads to the more gradual release of urea, helping plants gain more nutrients and resulting in higher yields.

16. What percentage of phosphorus (P_2O_5) and Nitrogen (N) is present in DAP ?

- (a) 10% N and 20% P_2O_5 (b) 18% N and 46% P_2O_5
(c) 30% N and 60% P_2O_5 (d) 35% N and 70% P_2O_5

U.P.P.C.S. (Mains) 2008

Ans. (b)

The Diammonium Phosphate (DAP) is used as a fertilizer. Its formula is $(\text{NH}_4)_2\text{HPO}_4$. The average pH in solution is 7.5–8. The typical formulation is 18% N and 46% P_2O_5 .

17. Which one of the following is a mixed fertilizer ?

- (a) Urea (b) CAM
(c) Ammonium Sulphate (d) NPK

I.A.S. (Pre) 1995

Ans. (d)

Among the given options, NPK is a mixed fertilizer. NPK fertilizer is a complex fertilizer comprised primarily of three main nutrients (Nitrogen, Phosphorous and Potassium) required for healthy plant growth. Different ratios of NPK in fertilizer serves different needs of soils.

18. Which of the following is not a chemical fertilizer?

- (a) Urea (b) Sodium Sulphate
(c) Super Phosphate (d) Potassium Nitrate

Uttarakhand P.C.S. (Pre) 2010

Ans. (b)

Urea, super phosphate and potassium nitrate are chemical fertilizers. Sodium sulphate Na_2SO_4 is the sodium salt of sulphuric acid. Sodium sulphate is not used as a chemical fertilizer.